



Universitatea
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Facultatea de
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Catedra de
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Smart Central Heating Controller

Distributed Local-Network IoT Thermostat System

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Objective

Retrofit a traditional central heating boiler with a local-network smart thermostat that can monitor room conditions, provide a simple web dashboard, and safely switch the boiler through a relay node.

- ▶ No cloud dependency.
- ▶ Two separate boards: one for sensing and control, one for relay actuation.
- ▶ Designed for practical home use and safe boiler integration.

2. Two-Node Architecture

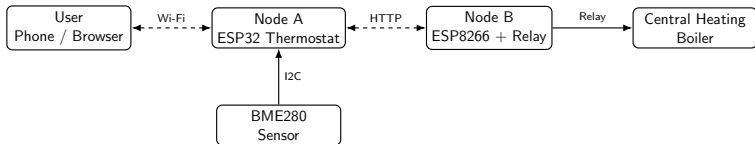
Node A: Thermostat

- ▶ ESP32 WROOM
- ▶ BME280 sensor
- ▶ Placed in the living area
- ▶ Reads temperature and humidity
- ▶ Hosts the web interface and control logic

Node B: Actuator

- ▶ ESP8266 board
- ▶ 5V relay module
- ▶ Placed near the boiler
- ▶ Receives commands from Node A
- ▶ Switches the boiler thermostat loop

3. System Diagram



4. Software & User Interface

- ▶ **Firmware:** C++ with Arduino Core in PlatformIO.
- ▶ **Frontend:** HTML, CSS, JavaScript with asynchronous `fetch()` requests.
- ▶ **Visualization:** Chart.js for recent temperature history.
- ▶ **Node A role:** reads the sensor, evaluates heating logic, and exposes the thermostat API.
- ▶ **Node B role:** exposes a minimal relay API for ON/OFF commands.
- ▶ **Data strategy:** recent measurements are kept in a 180-point RAM circular buffer to avoid unnecessary flash writes.

5. Safety, API & Conclusion

Safety

- ▶ Boiler switching is isolated to the relay node.
- ▶ The relay is used as a dry contact on the low-voltage thermostat loop.
- ▶ Node B can default to OFF if communication is lost.

Main API

- ▶ Node A: `/api/status`
- ▶ Node A: `/api/override`
- ▶ Node B: `/relay/on`
- ▶ Node B: `/relay/off`

Conclusion

The project delivers a simple and realistic smart thermostat architecture: the user interacts with the ESP32 thermostat over Wi-Fi, and the thermostat controls the boiler through a second board dedicated to the relay.